

Integrali s parametri

$a, b, c, d \in \mathbb{R}, f : [a, b] \times [c, d] \rightarrow \mathbb{R}, F(t) = \int_a^b f(x, t) \, dx$:

- f zvezna $\implies F$ zvezna.
- f zvezna $\implies \int_c^d (\int_a^b f(x, t) \, dx) \, dt = \int_a^b (\int_c^d f(x, t) \, dt) \, dx$.
- f zvezna, $\frac{\partial f}{\partial t}$ zvezen $\implies F$ odv., $\frac{dF}{dt} = \int_a^b \frac{\partial f}{\partial t}(x, t) \, dx$.

$a, b, c, d \in \mathbb{R}, f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ zvezna, $\frac{\partial f}{\partial t}$ zvezen,

$\alpha, \beta : [c, d] \rightarrow [a, b] \implies F(t) = \int_{\alpha(t)}^{\beta(t)} f(x, t) \, dx$ odvedljiva in

$$\frac{dF}{dt} = \int_{\alpha(t)}^{\beta(t)} \frac{\partial f}{\partial t}(x, t) \, dx + f(\beta(t), t) \frac{d\beta}{dt} - f(\alpha(t), t) \frac{d\alpha}{dt}.$$

$a, b, c \in \mathbb{R}, f : [a, \infty) \times [c, d] \rightarrow \mathbb{R}$ zvezna. $\int_a^\infty f(x, t) \, dx$ enakomerno konvergira na $[c, d]$, če $\forall \epsilon > 0 \exists M_0 \geq a : |\int_M^\infty f(x, t) \, dx| < \epsilon, \forall t \in [c, d], \forall M \geq M_0$.

$a, b, c \in \mathbb{R}, f : [a, \infty) \times [c, d] \rightarrow \mathbb{R}$ zvezna in $\exists$ integrabilna $\varphi : [a, \infty) \rightarrow \mathbb{R}_+ : |f(x, t)| \leq \varphi(x), \forall x \geq a, \forall t \in [c, d] \implies \int_a^\infty f(x, t) \, dx$ konvergira enakomerno.

$a, b, c \in \mathbb{R}, f : [a, \infty) \times [c, d] \rightarrow \mathbb{R}, F(t) = \int_a^\infty f(x, t) \, dx$:

- f zvezna in F konvergira enakomerno $\implies F$ zvezna.
- f zvezna in F konvergira enakomerno $\implies \int_c^d (\int_a^\infty f(x, t) \, dx) \, dt = \int_a^\infty (\int_c^d f(x, t) \, dt) \, dx$.
- f zvezna, $\frac{\partial f}{\partial t}$ zvezen, $F(t)$ konvergira $\forall t$ in $G(t) = \int_a^\infty \frac{\partial f}{\partial t}(x, t) \, dx$ konvergira enakomerno $\implies F$ odvedljiva in $\frac{dF}{dt} = G$.

[Fubini] $a, b, c, d \in \mathbb{R} \cup \{\pm\infty\}, f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ zvezna.

$f(x, y) \geq 0$ ali $\int_a^b \int_c^d |f(x, y)| \, dt \, dx < \infty \implies$

$$\int_a^b \int_c^d f(x, y) \, dt \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dt.$$

(f_n) konvergira enakomerno k f na vsakem omejenem intervalu, $\exists M : \int_{-\infty}^\infty f_n(x) \, dx$ konvergira enakomerno $\forall x > M$ in $\int_{-\infty}^\infty f(x) \, dx$ konvergira $\implies \lim_{n \rightarrow \infty} \int_{-\infty}^\infty f_n(x) \, dx = \int_{-\infty}^\infty f(x) \, dx$.

Eulerjevi Γ in B funkciji

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x} \, dx, \quad t > 0$$

$$\Gamma(t+1) = t\Gamma(t)$$

$$\Gamma(n) = (n-1)!, \forall n \in \mathbb{Z}^+$$

Γ je lokalno enakomerno konvergentna na $(0, \infty)$.

$$\Gamma \in C^\infty(0, \infty).$$

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} \, dx, \quad p, q > 0$$

$$B(p, q) = B(q, p)$$

$$B(\alpha, \beta) = 2 \int_0^{\frac{\pi}{2}} \sin^{2\alpha-1} x \cos^{2\beta-1} x \, dx, \quad \alpha, \beta > 0$$

$$B(p, q) = \int_0^\infty \frac{u^{p-1}}{(1+u)^{p+q}} \, du$$

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

$$B(p, 1-p) = \Gamma(p)\Gamma(1-p) = \frac{\pi}{\sin(p\pi)}$$

$$[\text{Stirling}] \lim_{x \rightarrow \infty} \frac{\Gamma(x+1)}{(\frac{x}{e})^x \sqrt{x}} = \sqrt{2\pi}$$

Večkratni Riemannov integral

Omejena funkcija je integrabilna, če je njen spodnji Darbouxov integral ($s(f)$) enak zgornjemu ($S(f)$).

f integrabilna $\iff$ obstaja limita $R(f, D, \xi)$.

$f, g : D \subset \mathbb{R}^n$ integrabilni, $c \in \mathbb{R}$:

- $\int_D (f+g) \, dV = \int_D f \, dV + \int_D g \, dV$
- $\int_D cf \, dV = c \int_D f \, dV$
- $\int_D |f| \, dV \geq |\int_D f \, dV|$
- $f \leq g : \int_D f \, dV \leq \int_D g \, dV$
- D ima prostornino: $|\int_D f \, dV| \leq \sup_D |f| V(D)$
- $D = A \cup B, V(A \cap B) = 0 : \int_D f \, dV = \int_A f \, dV + \int_B f \, dV$

Prostornina in mera

Omejena množica $A \subset \mathbb{R}^n$ ima n -razsežno prostornino, če je $f = 1$ integrabilna na A . Tedaj $V(A) = \int_A 1 \, dV$.

A ima prostornino $\iff V(\partial A) = 0$.

$V(A) = 0 \iff \forall \epsilon > 0 \exists$ kvadri $P_1, \dots, P_n : \cup P_j \supset A$ in $\sum V(P_j) < \epsilon$.

Končna unija množic s prostornino 0 ima prostornino 0.

$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ integrabilna $\implies V(G(f)) = 0$.

$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ integrabilna in $A \subset D, V(A) = 0$. Če $f|_A$ spremenimo za omejeno funkcijo, se njen integral ne spremeni.

$A \subset \mathbb{R}^n$ ima mero 0 ($m(A) = 0$), če $\forall \epsilon > 0 \exists$ števno mnogo kvadrov $A_j : A \subset \cup A_j, \sum V(A_j) < \epsilon$.

Števena unija množic z mero 0 ima mero 0.

A kompaktna in $m(A) = 0 \iff V(A) = 0$.

[Lebesgue] $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ omejena. f integrabilna na $D \iff$ množica nezveznosti ima mero 0.

Iterirani integral

$f : D = [a, b] \times [c, d] \rightarrow \mathbb{R}$ integrabilna in $f(x, \cdot) : [c, d] \rightarrow \mathbb{R}$ integr. $\forall x \in [a, b] \implies \int \int_D f(x, y) \, dx \, dy = \int_a^b \int_c^d f(x, y) \, dy \, dx$.

$f : D = [a, b] \times [c, d] \rightarrow \mathbb{R}$ zvezna $\implies$

$$\int \int_D f(x, y) \, dx \, dy = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy.$$

$\alpha, \beta : [a, b] \rightarrow \mathbb{R}$ zvezni, $f : A \rightarrow \mathbb{R}$ zvezna, $A = \{(x, y) : a \leq x \leq b, \alpha(x) \leq y \leq \beta(x)\} \implies \int \int_A f(x, y) \, dx \, dy = \int_a^b \int_{\alpha(x)}^{\beta(x)} f(x, y) \, dy \, dx$.

$f : D = D_2 \times [a, b] \rightarrow \mathbb{R}$ zvezna $\implies$

$$\int \int \int_D f \, dV = \int \int_{D_1} \int_a^b f \, dz \, dS.$$

Zamenjava spremenljivk

Jf je determinanta Jacobijeve matrike.

$A \subset \mathbb{R}^n$, odprta, omejena, s prostornino, $\varphi : A \rightarrow \mathbb{R}^n$ injektivna, $\varphi \in C^1(A), J\varphi \neq 0 \forall x \in A, J\varphi$ omejena na A . $\varphi(A)$ ima prostornino in je odprta ter $f : \varphi(A) \rightarrow \mathbb{R}$ integrabilna $\implies \int_{\varphi(A)} f(x) \, dx = \int_A f(\varphi(t)) |J\varphi(t)| \, dt$.

Polarne koordinate:

$$x = r \cos \varphi \quad y = r \sin \varphi$$

$$J = r$$

Cilindrične koordinate:

$$x = r \cos \varphi \quad y = r \sin \varphi \quad z = z$$

$$J = r$$

Sferične koordinate:

$$x = r \sin \vartheta \cos \varphi \quad y = r \sin \vartheta \sin \varphi \quad z = r \cos \vartheta$$

$$J = r^2 \sin \vartheta$$

Uporabna večkratnega integrala

$T \subset \mathbb{R}^3$ telo, $\rho = \rho(x, y, z)$ gostota.

$$m(T) = \iiint_T \rho \, dx \, dy \, dz \quad (\text{masa})$$

$$x_t = \frac{1}{m(T)} \iiint_T x \rho \, dx \, dy \, dz \quad (\text{težišče})$$

$$J_z = \iiint_T (x^2 + y^2) \rho \, dx \, dy \, dz \quad (\text{vztrajnostni moment})$$

Posplošeni integral

I. a) $f \geq 0, D \subset \mathbb{R}^n$ neomejena:

Definiramo $D_j = D \cap [-j, j]^n$, privzemimo $\forall j : \exists \int_{D_j} f(x) \, dx$.

Tedaj $\int_D f(x) \, dx = \lim_{j \rightarrow \infty} \int_{D_j} f(x) \, dx$.

I. b) Če je $f : D \rightarrow \mathbb{R}$, definiramo $f^+ = \max\{f, 0\}$ in $f^- = \max\{-f, 0\}$. Privzemimo $\forall j : \exists \int_{D_j} f^\pm(x) \, dx$.

Tedaj $\int_D f(x) \, dx = \int_{D_j} f^+(x) \, dx - \int_{D_j} f^-(x) \, dx$.

II. a) D omejena, $f : D \rightarrow (\mathbb{R}^+ \cup \{0\})$.

$$f_j = \begin{cases} f(x); & 0 \leq f(x) \leq j \\ j; & f(x) > j \end{cases}, \text{ privzemimo } \forall j : \exists \int_D f_j(x) \, dx.$$

Tedaj $\int_D f(x) \, dx = \lim_{j \rightarrow \infty} \int_D f_j(x) \, dx$.

II. b) Če je $f : D \rightarrow \mathbb{R}$.

Privzemimo $\forall j : \exists \int_D f_j^\pm(x) \, dx$.

Tedaj $\int_D f(x) \, dx = \int_D f_j^+(x) \, dx - \int_D f_j^-(x) \, dx$.

$f \geq 0 \implies$ obstoj in vrednost integrala f nista odvisna od izbire $\{D_j\}$.

$\exists \int_D |f(x)| \, dx \implies \exists \int_D f(x) \, dx$ in ni odvisen od izbire $\{D_j\}$.

Vektorska analiza

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

$$\operatorname{grad} u = \nabla u$$

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$\operatorname{rot} \mathbf{F} = \nabla \times \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

$$\Delta f = \nabla^2 f = \operatorname{div} \operatorname{grad} f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\operatorname{rot} \operatorname{grad} f = 0$$

$$\operatorname{div} \operatorname{rot} \mathbf{F} = 0$$

$$\nabla (fg) = g \nabla f + f \nabla g$$

$$\operatorname{div} (f\mathbf{F}) = f \operatorname{div} \mathbf{F} + \nabla f \cdot \mathbf{F}$$

$$\operatorname{div} (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot \operatorname{rot} \mathbf{F} - \mathbf{F} \cdot \operatorname{rot} \mathbf{G}$$

Vektorsko polje **F** je, potencialno, če $\exists u : D \rightarrow \mathbb{R} : \mathbf{F} = \nabla u$.

$\mathbf{F} = \nabla u$ in γ odsekoma gladka krivulja od A do $B \implies \int_\gamma \mathbf{F} \cdot d\mathbf{r} = u(B) - u(A)$.

F vektorsko polje na $K(s, R) \subset \mathbb{R}^3$ in $\operatorname{rot} \mathbf{F} = 0 \implies \mathbf{F}$ potencialno.

F vektorsko polje. Naslednje trditve so ekvivalentne:

- F** je potencialno.
- Integral **F** po vsaki sklenjeni krivulji je enak 0.
- Integral **F** je neodvisen od poti med dvema točkama.

Krivulje v prostoru

$$s(t) = \int_0^t \|\dot{\mathbf{r}}(\tau)\| d\tau$$

$\rho(s) = \mathbf{r}(t(s))$ je naravna parametrizacija

$$\mathbf{T} = \frac{\dot{\mathbf{r}}}{\|\dot{\mathbf{r}}\|} \quad (\text{enotski tangentni vektor})$$

$$\mathbf{B} = \frac{\dot{\mathbf{r}} \times \ddot{\mathbf{r}}}{\|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}\|} \quad (\text{enotski binormalni vektor})$$

$$\mathbf{N} = \mathbf{B} \times \mathbf{T} = \frac{(\dot{\mathbf{r}} \times \ddot{\mathbf{r}}) \times \dot{\mathbf{r}}}{\|(\dot{\mathbf{r}} \times \ddot{\mathbf{r}}) \times \dot{\mathbf{r}}\|} \quad (\text{enotski normalni vektor})$$

$$\kappa = \frac{1}{r_{\text{prit}}} = \left\| \frac{d\mathbf{T}}{ds} \right\| = \frac{\|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}\|}{\|\dot{\mathbf{r}}\|^3} \quad (\text{fleksijska ukrivljenost})$$

$$\omega = -\dot{\mathbf{B}} \cdot \mathbf{N} = \frac{(\dot{\mathbf{r}} \times \ddot{\mathbf{r}}) \cdot \ddot{\mathbf{r}}}{\|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}\|^2} \quad (\text{torzijska ukrivljenost})$$

B je normala pritisnjene ravnine.

T je normala normalne ravnine.

N je normala rektifikacijske ravnine in kaže v smeri središča pritisnjene krožnice.

$$\frac{d}{ds} \begin{bmatrix} T \\ N \\ B \end{bmatrix} = \begin{bmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \omega \\ 0 & -\omega & 0 \end{bmatrix} \begin{bmatrix} T \\ N \\ B \end{bmatrix} \quad (\text{Frenet-Serretove formule})$$

Orientacija je zvezen izbor enotskega tangentnega vektorja.

Ploskve v prostoru

Zahtevamo $\mathbf{r} \in C^1$ in $\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \neq 0$.

$$\mathbf{N} = \frac{\mathbf{r}_u \times \mathbf{r}_v}{\|\mathbf{r}_u \times \mathbf{r}_v\|} \quad (\text{normala tangentne ravnine})$$

Če je ploskev podana z $F(x, y, z) = 0$, je normala ∇F .

$$\operatorname{pl}(\mathbf{r}(D)) = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| du dv$$

$$E = \|\mathbf{r}_u\|^2, G = \|\mathbf{r}_v\|^2, F = \mathbf{r}_u \cdot \mathbf{r}_v \quad (\text{koeficienti 1. fund. forme})$$

$$\|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{EG - F^2}$$

Orientacija je zvezen izbor enotskega normalnega vektorja.

Krivuljni in ploskovni integral

$$\int_\gamma f ds = \int_\alpha^\beta f(\mathbf{r}(t)) \|\dot{\mathbf{r}}(t)\| dt \quad (\text{skalarno polje})$$

$$\int_\gamma \mathbf{F} \cdot d\mathbf{r} = \int_\alpha^\beta \mathbf{F}(\mathbf{r}(t)) \cdot \dot{\mathbf{r}}(t) dt \quad (\text{vektorsko polje})$$

$$\int_{-\gamma} \mathbf{F} \cdot d\mathbf{r} = - \int_\gamma \mathbf{F} \cdot d\mathbf{r}$$

$$\int_\Sigma f dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv \quad (\text{skalarno polje})$$

$$\int_\Sigma \mathbf{F} dS = \iint_\Sigma \mathbf{F} \cdot \mathbf{N} dS = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv$$

Integralski izreki

$$\bar{\Omega} = \Omega \cup \partial\Omega$$

Gaussov izrek:

$\Omega \subset \mathbb{R}^3$ odprta omejena množica z odsekoma gladkim robom ($\partial\Omega$ odsekoma gladka ploskev z orientacijo, ki kaže iz Ω).

$\mathbf{F} \in C^1$ vektorsko polje definirano na neki okolici $\bar{\Omega}$.

Tedaj je $\iint_{\partial\Omega} \mathbf{F} dS = \iiint_\Omega \operatorname{div} \mathbf{F} dV$.

Greenova formula:

$D \subset \mathbb{R}^2$ omejena odprta množica z odsekoma gladkim robom

(∂D je končna unija odsekoma gladkih krivulj s pozitivno orientacijo). $\mathbf{F} = (X, Y) \in C^1$ vektorsko polje definirano na neki okolici $\bar{D}$.

Tedaj je $\int_{\partial D} (X dx + Y dy) = \iint_D (Y_x - X_y) dx dy$.

Stokesov izrek:

$\Sigma \subset \mathbb{R}^3$ omejena odsekoma gladka orientirana ploskev z odsekoma gladkim robom ($\partial\Sigma$ je končna unija odsekoma gladkih krivulj orientiranih skladno s Σ). $\mathbf{F} \in C^1$ vektorsko polje definirano v neki okolici $\bar{\Sigma}$.

Tedaj je $\int_{\partial\Sigma} \mathbf{F} \cdot d\mathbf{r} = \iint_\Sigma \operatorname{rot} \mathbf{F} dS$.

Drugi Greenovi formuli:

$\Omega \subset \mathbb{R}^3$ odprta omejena množica z odsekoma gladkim robom ($\partial\Omega$ odsekoma gladka ploskev z orientacijo, ki kaže iz Ω). u, v gladki skalarni polji definirani na neki okolici $\bar{\Omega}$. Tedaj:

$$1. \quad \iint_{\partial\Omega} u \frac{\partial v}{\partial \mathbf{n}} dS = \iiint_\Omega u \Delta v dV + \iiint_\Omega \nabla u \cdot \nabla v dV.$$

$$2. \quad \iint_{\partial\Omega} (u \frac{\partial v}{\partial \mathbf{n}} - v \frac{\partial u}{\partial \mathbf{n}}) dS = \iiint_\Omega (u \Delta v - v \Delta u) dV.$$

Random ostale zadeve

Trigonometrične formule

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$$

$$\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$$

$$\sin x \cos y = \frac{1}{2} (\sin(x+y) + \sin(x-y))$$

$$\cos x \cos y = \frac{1}{2} (\cos(x+y) + \cos(x-y))$$

$$\sin x \sin y = \frac{1}{2} (\cos(x-y) - \cos(x+y))$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

Integrali iracionalnih funkcij

$$\int \frac{dx}{\sqrt{x^2+px+q}} = \ln \left| x + \frac{p}{2} + \sqrt{x^2+px+q} \right| + C$$

$$\int \frac{dx}{\sqrt{-x^2+px+q}} = \arcsin \left(\frac{2x-p}{\sqrt{p^2+4q}} \right) + C$$

$$\int \frac{P(x)dx}{\sqrt{ax^2+bx+c}} = T(x)\sqrt{ax^2+bx+c} + K \int \frac{dx}{\sqrt{ax^2+bx+c}};$$

($a \neq 0$, P in T polinoma)

Univerzalna trigonometrična substitucija

$$tg \frac{x}{2} = t, dx = \frac{2dt}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, \sin x = \frac{2t}{1+t^2}$$

Odvodi

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \log_a(x) = \frac{1}{x \ln a}$$

Per partes

$$\int u dv = uv - \int v du$$